

# Product Manual

## Rectifier Board Bally / Stern

P/N TPP-1025



Document Revision – 1.2

### Description

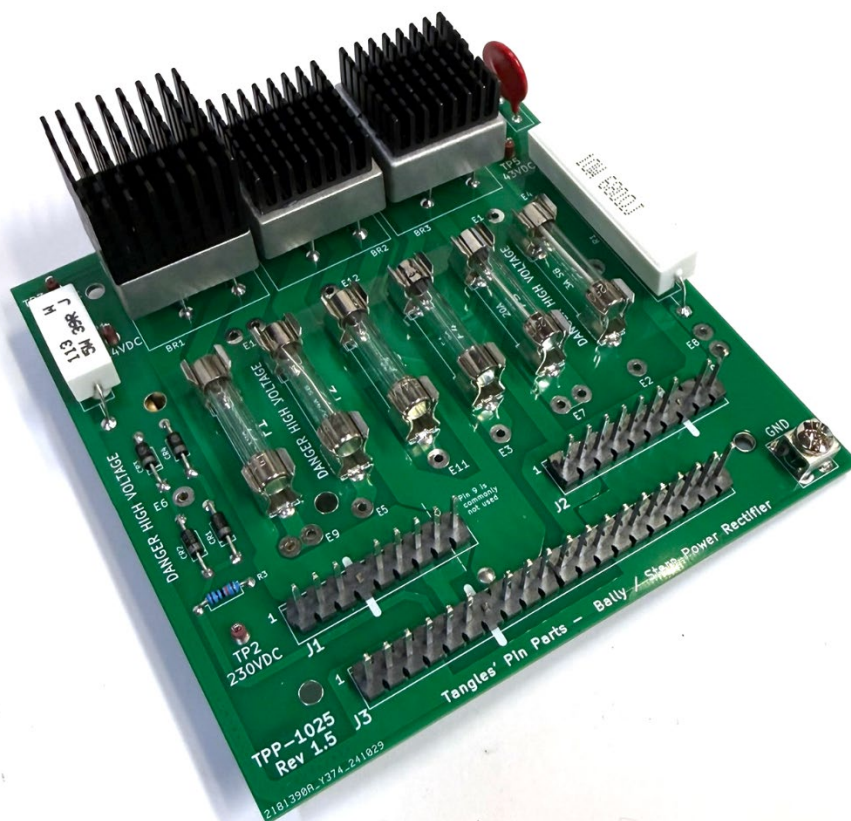
Tangles presents a board to replace the original Stern TA100 and Bally AS-2518-18 rectifier boards. The original boards suffer from undersized rectifiers, over-heating and high current stress in the through-hole plating of some connectors.

With the original weaknesses addressed in this new design, we offer:

- Three high current rectifiers with heatsinks.
- Quality Molex-branded connectors.
- Molex Trifurcon™ pins for extra current capabilities
- All high-current paths have expanded traces.
- Ceramic stand-off mounts for the high-power resistors.
- Screw terminal for enhanced grounding.
- Multiple solder points for transformer pins E7, E8, E9 and E10 for machines that have multiple wires at these points.
- 9th pin J1 - compatible with machines that need the extra wire.
- High-quality fuses used.
- **Poly-carbonate safety shield included.**

**Warning: Lethal voltages are present in Pinball machines and on this board. This service procedure is suitable for qualified persons only. A moderate level of servicing skill is required to complete the installation of this board.**

**NOTE: This board is suitable for experienced repairers only**



## Machine Compatibility

### Bally Games:

Black Jack, Dolly Parton, Eight Ball, Evel Knievel, Freedom, Frontier, Future Spa, Harlem Globetrotters, Hotdoggin', KISS, Lost World, Mata Hari, Mystic, Night Rider, Nitro Groundshaker, Paragon, Playboy, Power Play, Rolling Stones, Silverball Mania, Six Million Dollar Man, Skateball, Space Invaders, Star Trek, Strikes & Spares, Supersonic, Viking, Voltan

### Stern Games:

Ali, Big Game, Catacomb, Cheetah, Dracula, Dragonfist, Flight 2000, Free Fall, Galaxy, Hot Hand, Iron Maiden, Lectronamo, Lightning, Magic, Memory Lane, Meteor, Nineball, Orbitor One, Pinball, Quicksilver, Seawitch, Stargazer, Stars, Stingray, Split Second, Ted Nugent, Trident, Viper, Wildfyre

**Note: J1 Connector header pin 9 is used for KISS, Future Spa and Space Invaders. All other Bally and Stern games do not use pin 9.**

## Installation Notes

This document assumes those performing the installation will have a moderate level of skill in Pinball servicing to achieve a safe and effective installation.

Tools required:

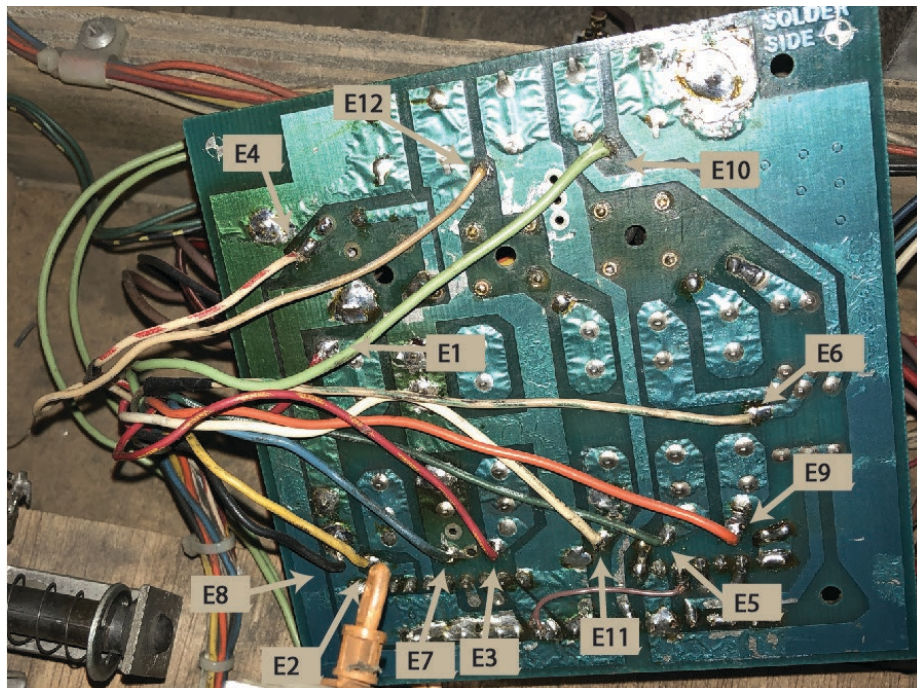
- Moderate to high-powered soldering iron.
- Multimeter (to verify voltages)
- Wire strippers
- Wire cutters
- Molex crimper suitable for crimping the .156" pins

When installing this board, it is important to identify the transformer wires soldered to the PCB's back. I present a table of standard colours that may assist with wire identification. I encourage installers to label each wire as they remove it from the old board so it does not get mixed up.

|     |               |                                                                                      |
|-----|---------------|--------------------------------------------------------------------------------------|
| E1  | Red           |  |
| E2  | Yellow        |  |
| E3  | Red / Yellow  |  |
| E4  | White / Red   |  |
| E5  | Green         |  |
| E6  | White / Green |  |
| E7  | Blue          |  |
| E8  | Black         |  |
| E9  | Orange        |  |
| E10 | Green         |  |
| E11 | White         |  |
| E12 | White / Black |  |

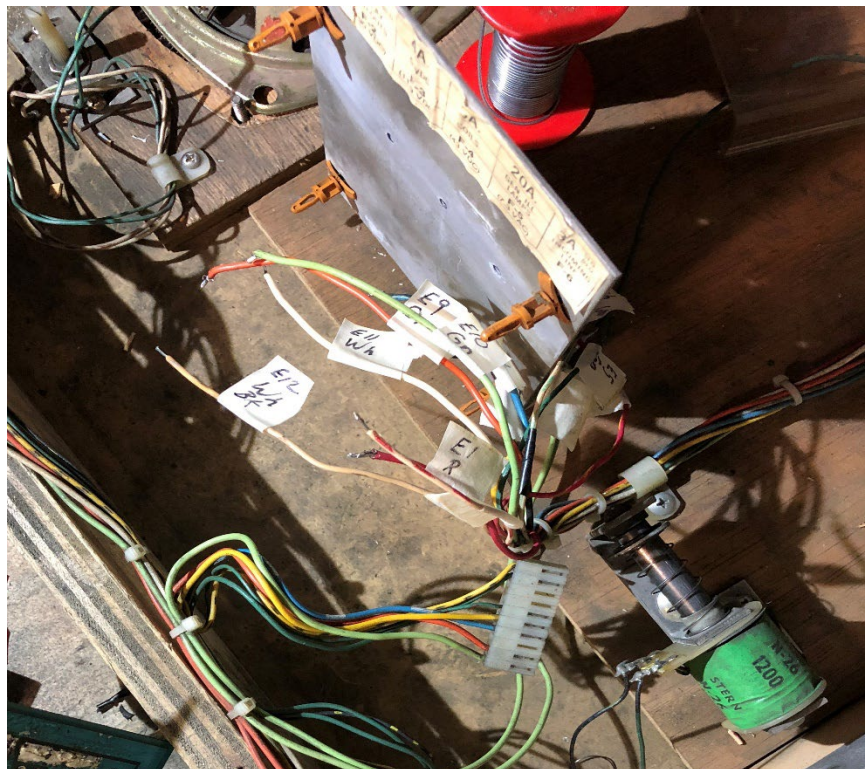
The wire colours provided should be taken as a guide only, as machines may have differences, or may have had wires replaced with non-standard colours.

Here's a photo of the underside of a TA100 board with the transformer points identified:



#### Part 1: Remove the old board

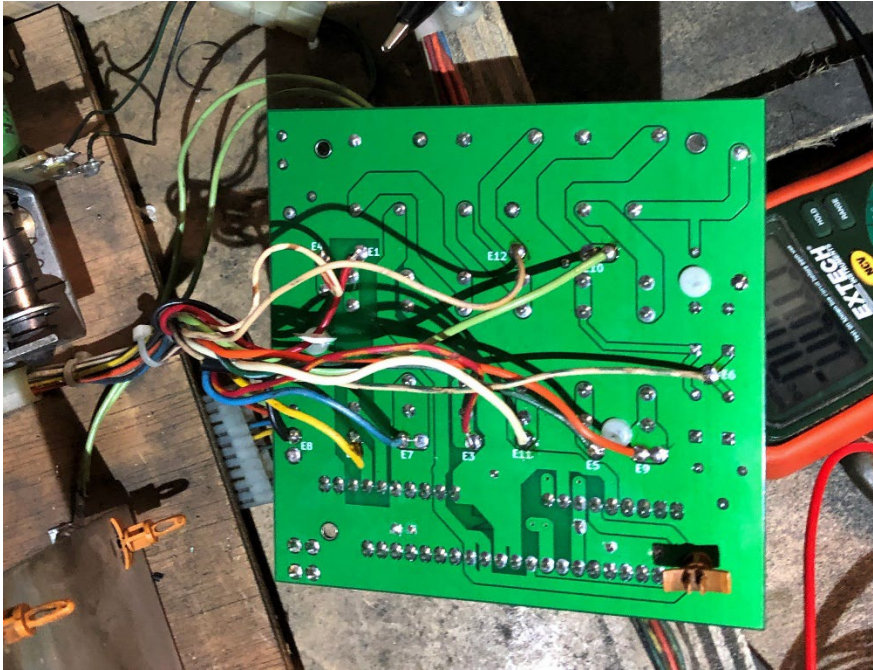
Remove the wires from the old board by desoldering or snipping one wire at a time. Attach a label to each wire using masking tape, identifying the E-location the wire came from. This ensures the wires are not mixed up in the transition.



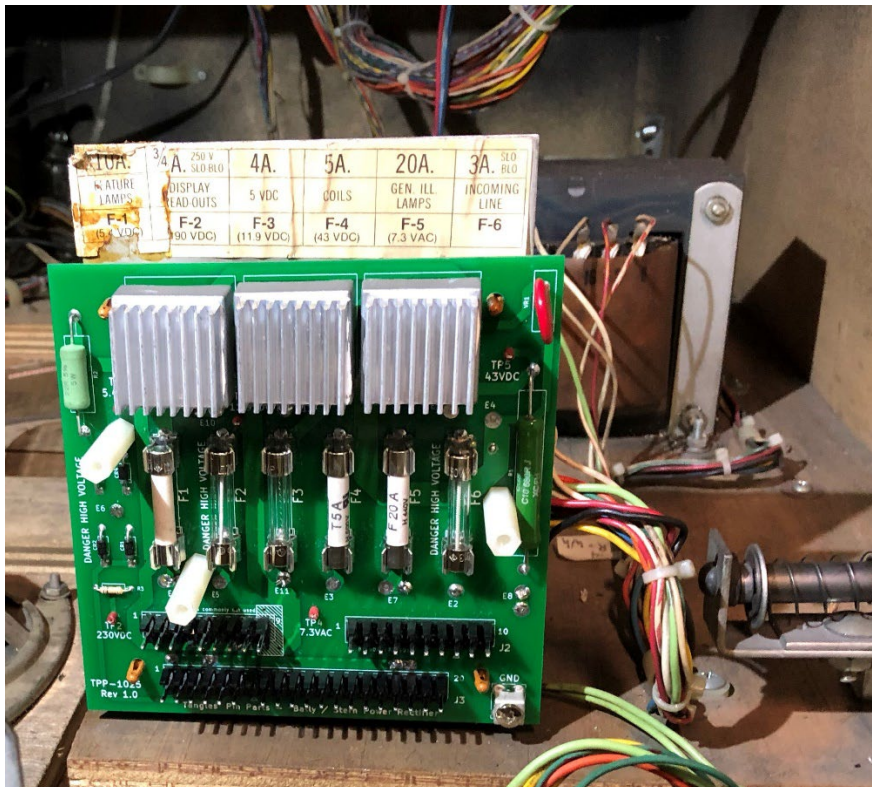
#### Part 2: Install the new board

After removing all wires from the old board, reconnect them to the new board. Note the new TPP-1025 board has the 'E' numbers printed on the reverse of the board.

It is recommended that the wires be inserted through the holes in the PCB and soldered on both the front and the back of the board.



Reattach the board to the mounting posts and install the Lexan safety shield.



### Part 3: Replace Plugs and Pins

If replacing the Molex connectors and pins on the wiring harness (recommended) I recommend this be done one wire at a time. This reduces the possibility of mixing the wires up.

Snip each wire in turn and:

- Strip the insulation
- Crimp on the Molex contact
- Insert into the Molex housing.

Doing one wire at a time means mistakes are less likely.

#### Part 4: Test and verify voltages



**Beware of the lethal voltages that are present on the board. Never touch the board when power is applied.**

- Ensure mains power is OFF
- Connect ONLY plug J2. (Plugs J1 and J3 disconnected)
- Connect the multimeter to read the appropriate test point and the ground pin.
- Turn the power on and note the voltage.
- Turn the power off to move the test leads to the next point.

I recommend using a multimeter with alligator clips.

## Fuses

Ensure power is off, and the unit is unplugged before changing fuses.

Standard fuses (supplied) are listed here. You need to check with your machine's manual for the correct fuse values.

|                     |                                        |
|---------------------|----------------------------------------|
| F1 = 10A            | 5.4V Feature Lamps                     |
| F2 = 3/4A Slow-Blow | 230VDC Legacy Display Power            |
| F3 = 4A             | 12V Unregulated - (feeds 5V regulator) |
| F4 = 5A             | 43V Solenoid power                     |
| F5 = 20A            | General Illumination                   |
| F6 = 3A Slow-Blow   | Mains (240V)                           |

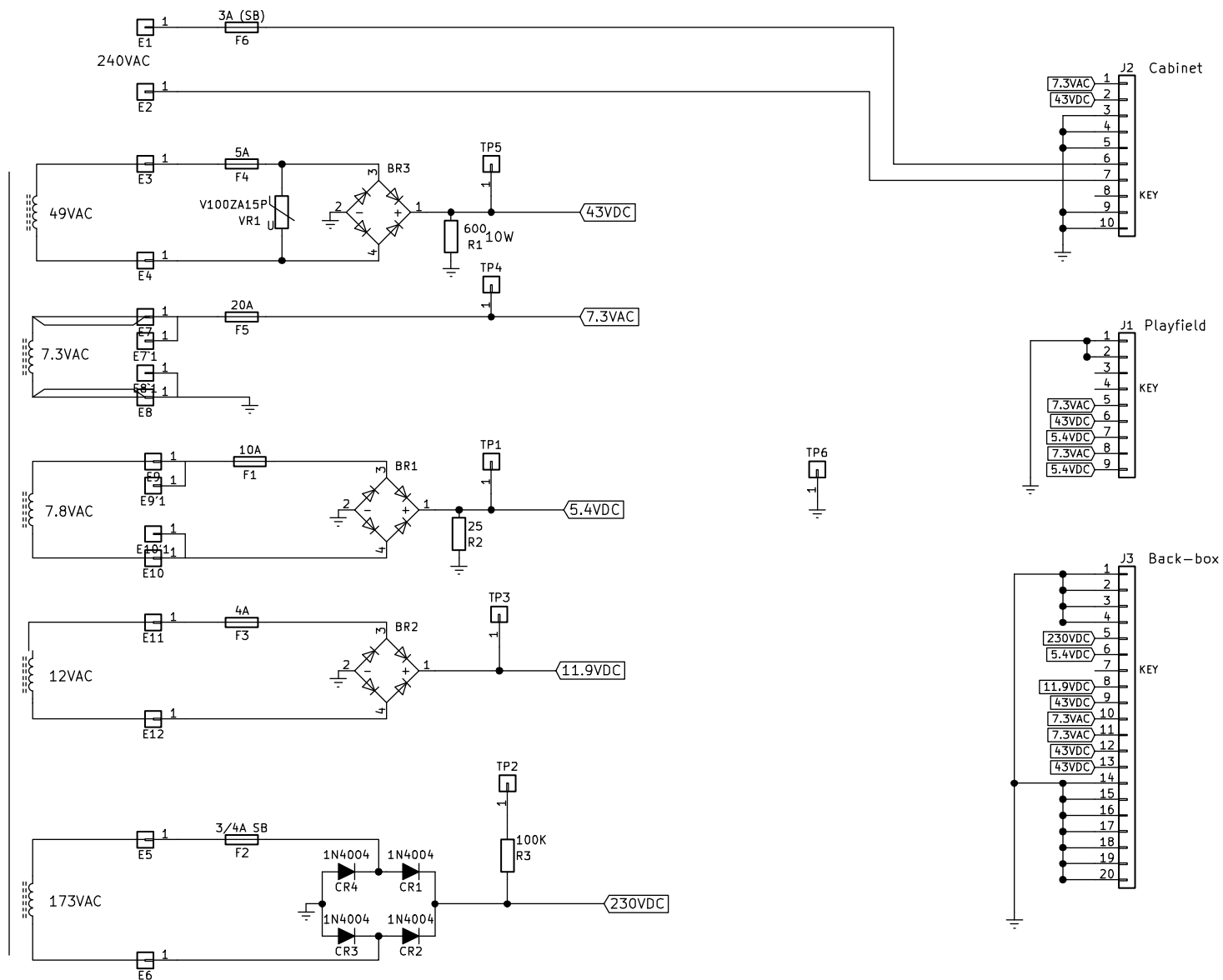
### LED Lamps Fitted

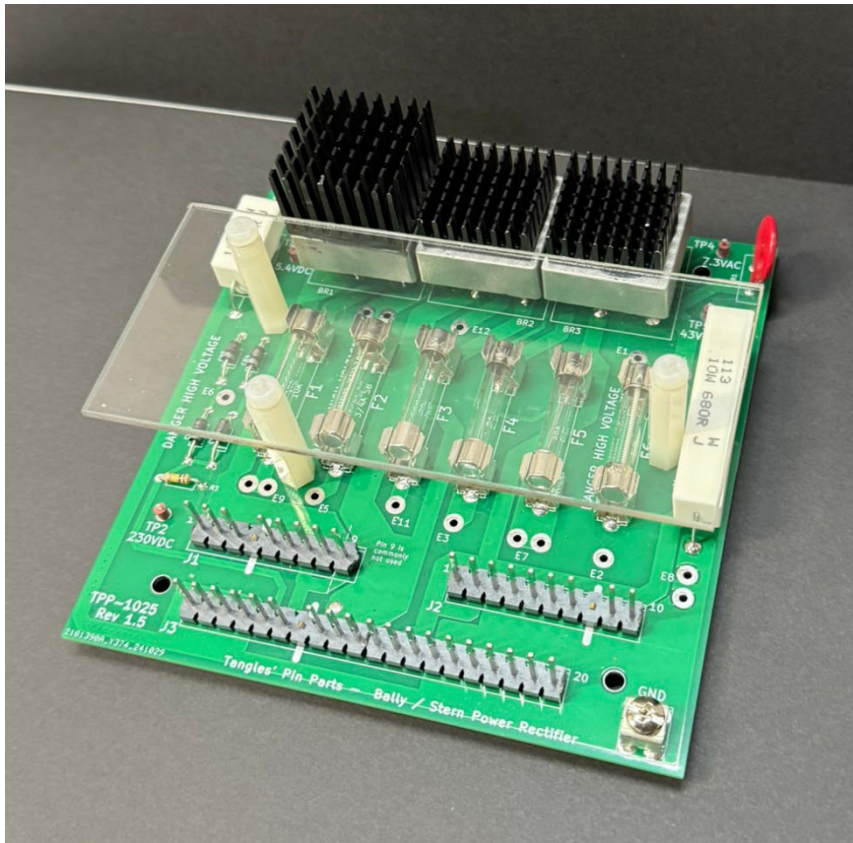
|         |                                    |
|---------|------------------------------------|
| F1 = 3A | 5.4V Feature Lamps – LEDs fitted.  |
| F5 = 4A | General Illumination – LEDs fitted |

### LED Displays Fitted

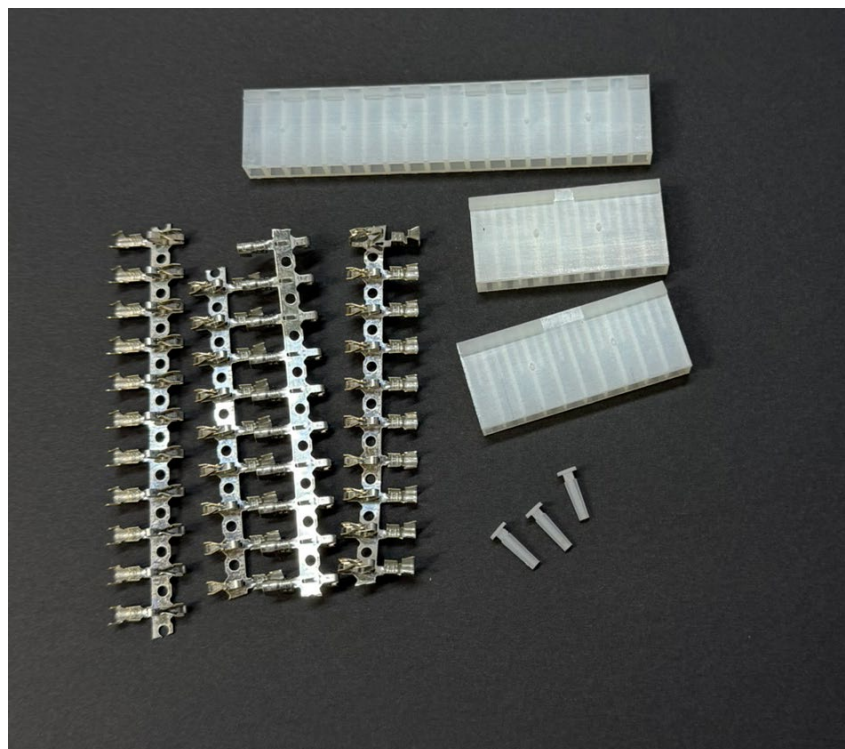
|          |                               |
|----------|-------------------------------|
| F2 = N/A | If using LED Display modules. |
|----------|-------------------------------|

*SIBA and Littelfuse branded fuses are recommended.*

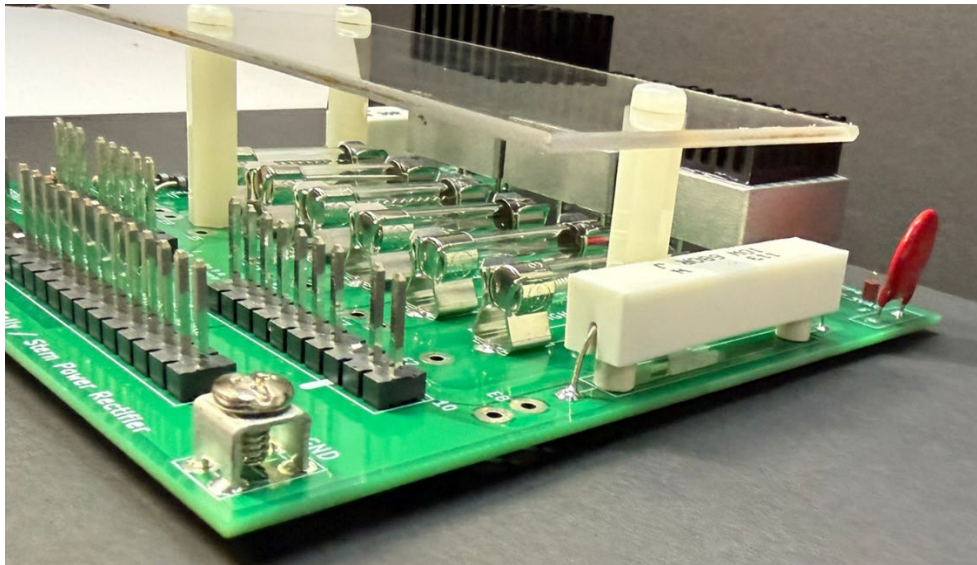




Polycarbonate safety shield installed.



Included Molex pins, shells and blanking pins.



Showing power resistors with ceramic stand-off.